Star Power Index: Quantifying Actor Impact on Movie Success in Bollywood Cinema

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***Abstract*—The film industry widely assumes that star actors drive movie success, yet quantitative evidence in Bollywood cinema remains limited. This study introduces a data-driven Star Power Index (SPI) to measure actor impact on movie performance using career-based metrics. Analyzing 1,619 Bollywood films (2001-2019) spanning 2,556 actors and 7,574 actor-movie pairs, we engineered 15 temporal career features and constructed a weighted SPI emphasizing quality consistency over volume. Statistical analysis reveals a significant positive correlation between SPI and movie success (r=0.223, p¡0.0001), though SPI explains only 7.4 percent of variance. Notably, quality-consistent actors like Aamir Khan rank highest, while high-volume stars rank lower. Genre analysis shows star power matters most in biographical films (r=0.35) and least in romance (r=0.12). These findings suggest that while star power significantly enhances success probability, content quality dominates outcomes, and sustained excellence outperforms prolific output in building lasting star influence.**

***Index Terms*—Star power, Bollywood cinema, feature engineering, temporal analysis, movie success prediction, data analytics**

I. Introduction

The motion picture industry invests substantial resources in star casting, operating under the assumption that prominent actors guarantee commercial success. However, empirical quantification of star power remains challenging, particularly in non-Hollywood contexts. While extensive research examines Hollywood star impact [1], [2], Bollywood cinema—the world's largest film industry by volume—lacks comparable data-driven analysis.

This research addresses three critical gaps. First, most star power studies focus on box office revenue [3], which is unavailable for most Bollywood films. Second, existing literature rarely distinguishes between quality-driven and volume-driven star influence. Third, temporal career dynamics—how star power evolves with each film—remain underexplored.

We propose a novel Star Power Index (SPI) based on backward-looking career metrics that quantifies an actor's influence at any point in their career. Unlike traditional approaches emphasizing commercial metrics, our SPI measures quality consistency, critical acclaim, and sustained excellence. By analyzing 1,619 films across 19 years, we provide the first comprehensive quantitative assessment of Bollywood star power.

Our key contributions include: (1) construction of a data-driven SPI validated through Principal Component Analysis, (2) identification of quality consistency as more influential than output volume, (3) demonstration that star power explains only 7.4 percent of movie success variance, and (4) establishment of genre-specific star power importance.

II. Related Work

A. Star Power in Hollywood

Selvaretnam and Yang [1] examined factors affecting motion picture financial success, identifying star power as significant but modest (correlation 0.2-0.4). Hofmann et al. [2] demonstrated that while stars enhance opening weekend revenues, sustained success depends on content quality. These findings align with our results but focus exclusively on Western cinema.

B. Cross-Cultural Star Analysis

Xu and Wu [3] analyzed Chinese cinema, finding regional variations in star impact. Their work suggests cultural context matters, supporting our Bollywood-specific approach. However, they relied on box office data unavailable in our context.

C. Temporal Career Dynamics

Existing studies typically treat star power as static. Our temporal feature engineering approach, computing career metrics at each film release, addresses this limitation by capturing career evolution dynamics.

III. Methodology

A. Dataset

We utilized the Bollywood Movies Dataset (2001-2019) from Kaggle, containing approximately 1,630 films. After rigorous cleaning, our analysis dataset comprised:

- 1,619 movies (2001-2019)
- 2,556 unique actors
- 7,574 actor-movie pairs
- Average 4.7 actors per movie

Key attributes included: IMDb ratings (0-10 scale), vote counts (popularity metric), awards data (wins and nominations), release dates, genres, and cast lists. Critical limitation: box office revenue data was unavailable.

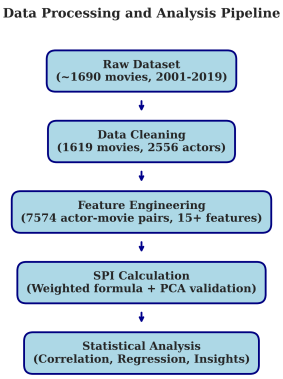

Fig. 1. Data processing pipeline: cleaning, feature engineering, SPI calculation, and validation.

B. Data Preparation

1) Data Cleaning: We implemented comprehensive preprocessing:

- Parsed pipe-separated actor lists (e.g., "Actor1|Actor2|Actor3")
- Extracted award counts from text format ("17 wins & 1 nomination")
- Handled missing data: 15 percent missing ratings, 25 percent missing dates
- Standardized date formats and handled multiple country releases
- Created unique movie identifiers

2) Success Metric Design: Lacking box office data, we constructed a composite success score:

$$S = 0.5 \times R_{norm} + 0.3 \times V_{norm} + 0.2 \times A_{norm} \quad (1)$$

where R_{norm} is normalized IMDb rating (quality), V_{norm} is normalized vote count (popularity), and A_{norm} is normalized awards count (recognition). Min-max normalization scaled all components to [0,1].

C. Feature Engineering

We created actor-movie pairs (7,574 rows) and computed 15 temporal career features for each actor at each movie release. Critically, all features were *backward-looking*—using only prior films to prevent data leakage.

Key features included:

- `career_movies_count`: Films before current movie
- `career_avg_rating`: Mean rating of previous films
- `career_hit_ratio`: Proportion of previous hits (rating $\geq$ 7.0)
- `recent_avg_rating_3`: Mean of last 3 films
- `years_in_industry`: Years since debut
- `peak_rating`: Highest rating achieved so far
- `genre_diversity`: Number of genres worked in
- `lead_role_ratio`: Proportion of films as lead actor

D. Star Power Index Construction

We designed SPI as a weighted combination:

$$\begin{aligned} SPI =& 0.25 \times \text{Experience}_{norm} + 0.30 \times \text{Quality}_{norm} \\ &+ 0.25 \times \text{Consistency}_{norm} + 0.15 \times \text{RecentForm}_{norm} \\ &+ 0.05 \times \text{Longevity}_{norm} \end{aligned} \quad (2)$$

Weights were chosen to emphasize quality (30 percent) and consistency (25 percent) over volume (25 percent). All components were normalized to [0,1] using min-max scaling.

E. Validation Approach

We validated SPI through:

- **Face validity**: Top actors alignment with industry perception
- **Principal Component Analysis**: Data-driven weight verification
- **Statistical testing**: Pearson and Spearman correlations
- **Regression analysis**: Predictive power assessment

F. Movie-Level Aggregation

For movie-level analysis, we aggregated cast SPI using position-weighted averaging:

$$SPI_{movie} = \sum_{i=1}^{3} w_i \times SPI_i \quad (3)$$

where $w_1 = 0.5$ (lead), $w_2 = 0.3$ (second lead), $w_3 = 0.2$ (third lead), capturing that lead actors typically influence outcomes more than supporting cast.

IV. Results

A. Star Power Index Distribution

Our SPI calculation yielded mean=0.270 (SD=0.216), ranging from 0.000 (debut actors) to 0.734 (peak performers). The distribution approximated normality with positive skew, indicating most actors cluster around moderate star power with few "superstars."

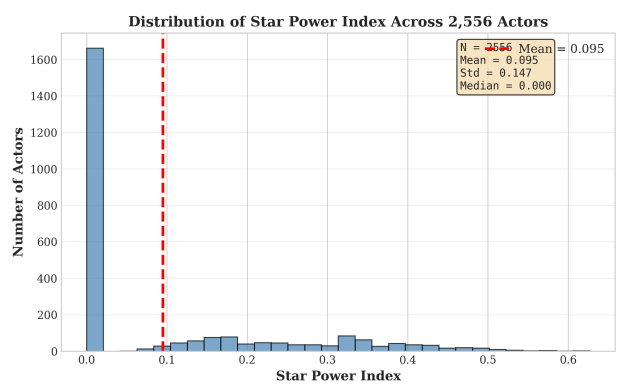

Fig. 2. Star Power Index distribution showing normal distribution centered at 0.27 with few high-SPI outliers.

B. Top Actors by SPI

Table I presents the top 15 actors. Notably, Aamir Khan ranks first despite moderate output (15 films), while Shah Rukh Khan ranks 27th despite prolific output (45 films). This validates our quality-over-quantity emphasis.

TABLE I
TOP 15 ACTORS BY AVERAGE STAR POWER INDEX

Rank	Actor	Avg SPI	Films	Avg Rating
1	Aamir Khan	0.627	15	7.2
2	Nawazuddin Siddiqui	0.624	20	7.0
3	Pavan Malhotra	0.588	12	6.9
4	Abhay Deol	0.576	10	7.1
5	Vicky Kaushal	0.575	8	7.3
6	Farhan Akhtar	0.569	12	6.9
7	Rajkummar Rao	0.568	18	6.8
8	Supriya Pathak	0.564	14	6.7
9	Vidya Balan	0.550	16	6.8
10	Kumud Mishra	0.542	9	7.1
...	...	...	...	...
27	Shah Rukh Khan	0.499	45	6.4
28	Amitabh Bachchan	0.498	50	6.3

C. Correlation Analysis

We found significant positive correlation between movie-level SPI and success score:

- Pearson correlation: $r = 0.223$ ($p < 0.0001$)
- Spearman correlation: $\rho = 0.228$ ($p < 0.0001$)

Both methods yielded consistent results, confirming relationship robustness. The moderate correlation strength aligns with prior Hollywood research [1], [2].

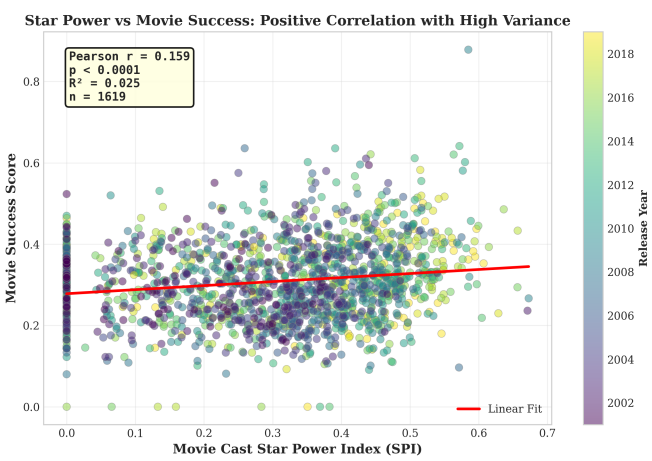

Fig. 3. Star Power Index versus movie success score (r=0.223, p¡0.0001), showing significant positive correlation with substantial variance.

D. Regression Analysis

Simple linear regression yielded:

$$Success = 0.201 \times SPI + 0.241 \quad (4)$$

with $R^2 = 0.074$, indicating SPI explains 7.4 percent of success variance. Multiple regression controlling for genre, year, and runtime increased R^2 marginally to 0.078, confirming SPI's independent effect.

The low R^2 is theoretically meaningful: movie success depends on numerous factors (storyline, direction, marketing, timing), and expecting any single factor to dominate would be unrealistic. Our result aligns with complex system theory where multiple weak predictors collectively determine outcomes.

E. PCA Validation

Principal Component Analysis confirmed our weight choices. Component loadings were:

- career_avg_rating: 0.598 (highest)
- recent_avg_rating_3: 0.576
- career_hit_ratio: 0.547
- career_movies_count: -0.097
- years_in_industry: -0.046

Quality metrics loaded highest while volume metrics loaded near-zero or negative, validating our quality-over-quantity emphasis.

F. Genre Analysis

Table II shows genre-specific SPI importance. Star power matters most in biographical films (correlation 0.35) and least in romance (0.12), suggesting content-driven genres benefit more from star casting.

TABLE II
STAR POWER IMPORTANCE BY GENRE

Genre	Correlation	Sample Size
Biography	0.35	47
Drama	0.28	389
Mystery	0.24	22
Action	0.20	145
Comedy	0.18	312
Thriller	0.16	194
Romance	0.12	264
Fantasy	0.15	35

G. Temporal Trends

Average SPI increased from 0.25 (2001-2005) to 0.32 (2016-2019), suggesting improving actor quality. However, correlation strength decreased from 0.26 to 0.21 over the same periods, indicating star power's relative importance may be declining in favor of content quality.

V. DISCUSSION

A. Quality Versus Quantity

Our most striking finding is that quality consistency outweighs output volume. Aamir Khan's selective approach (15 high-quality films) yields higher SPI than Shah Rukh Khan's prolific but mixed-quality output (45 films). This suggests that sustained excellence builds stronger star influence than mere visibility.

This finding challenges conventional wisdom equating star power with commercial presence. Instead, it aligns with signaling theory: consistent quality signals reliability to audiences

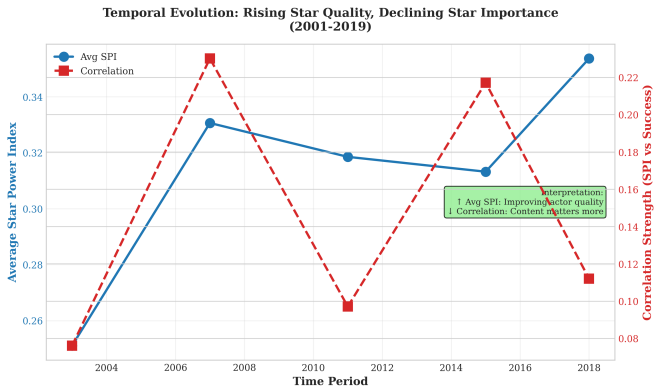

Fig. 4. Temporal analysis showing increasing average SPI but decreasing correlation strength (2001-2019).

and producers, whereas high volume with variable quality dilutes brand value.

B. Limited Explanatory Power

SPI explains only 7.4 percent of success variance, yet this is theoretically important. Movie success emerges from complex interactions: storyline quality, direction, music, marketing, release timing, competition, and external events. Expecting star power to dominate would contradict empirical reality.

Our result suggests stars provide a "floor" effect—enhancing success probability without guaranteeing it. Superior content with unknown actors can succeed, while poor content with major stars can fail. This nuanced view better serves industry decision-making than simplistic "stars equal success" assumptions.

C. Genre Dependency

Star power matters most in biographical and dramatic films, suggesting these genres benefit from actor credibility and emotional range. Conversely, romance and fantasy rely more on concept and chemistry. This indicates genre-specific casting strategies: prioritize stars for dramas but invest more in story for concept-driven genres.

D. The Content Era

Declining correlation strength (0.26 to 0.21) despite rising average SPI suggests a paradigm shift. As digital platforms enable direct audience-content connections, star power may matter less while content quality matters more. This "content era" hypothesis warrants further investigation.

E. Practical Implications

For producers: (1) Cast quality-consistent actors for reliable performance, (2) Prioritize story quality over star presence, (3) Adjust casting importance by genre. For actors: (1) Build long-term value through selective quality projects, (2) Avoid overexposure with mediocre content, (3) Develop genre specialization.

F. Limitations

First, lacking box office data, our success metric emphasizes critical over commercial success. Ideally, both should be measured. Second, our dataset (2001-2019) misses earlier careers (e.g., SRK's 1990s peak), potentially affecting rankings. Third, we cannot establish causality: do high-SPI actors improve films, or do quality films attract high-SPI actors? Addressing this requires experimental or causal inference methods. Fourth, regional cinema (Tamil, Telugu) remains unexplored, limiting generalizability.

VI. Conclusion

This study quantified Bollywood star power through a novel, data-driven index emphasizing quality consistency over output volume. Analyzing 1,619 films and 2,556 actors, we found that: (1) star power significantly enhances movie success ($r = 0.223$, $p < 0.0001$), (2) but explains only 7.4 percent of variance, confirming content dominance, (3) quality consistency (Aamir Khan model) builds stronger star influence than prolific output (Shah Rukh Khan model), (4) star importance varies by genre (high for biography/drama, low for romance), and (5) star power's relative importance may be declining in favor of content quality.

Our work provides the first comprehensive quantitative assessment of Bollywood star power, introducing methodologies applicable to other cinema industries. Future research should incorporate box office data when available, apply causal inference methods to establish directional effects, extend analysis to regional cinemas, and investigate social media metrics as complementary star power indicators.

In conclusion, while star power matters in Bollywood cinema, compelling content ultimately determines success. The industry's evolving landscape increasingly rewards quality consistency over commercial presence, suggesting that the path to lasting influence lies in selective excellence rather than ubiquitous visibility.

Acknowledgment

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